



RME 4201: Human Robot Interaction

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Course content

Introduction

- Overview of Human-Robot Interaction, Difference between Human Robot Interaction (HRI), Robotics and Human Computer Interaction (HCI), Evolution of HRI, HRI Taxonomies, Application of HRI.

Design

- Design in HRI, Antropomorphization in HRI, Design methods, Culture in HRI.

Spatial interaction

- Role of Space in Human-Robot Interaction, Spatial Interaction for Robots.

Trajectory Generation

- Mechanical manipulators, Trajectory generation using cubic and higher order polynomials.

Course content

Verbal Interaction

- Speech Recognition, Dialogue Management and Speech Production for HRI.

Nonverbal Interaction in HRI

- Categories of Nonverbal Interaction, Importance of Nonverbal Interaction, How Robot Perceive Nonverbal Cues, and How Robot Can Generate Nonverbal Cues during Interaction

Emotion in HRI

- Expression and Perception of Emotion HRI, Models of Emotions

Research Methods in HRI

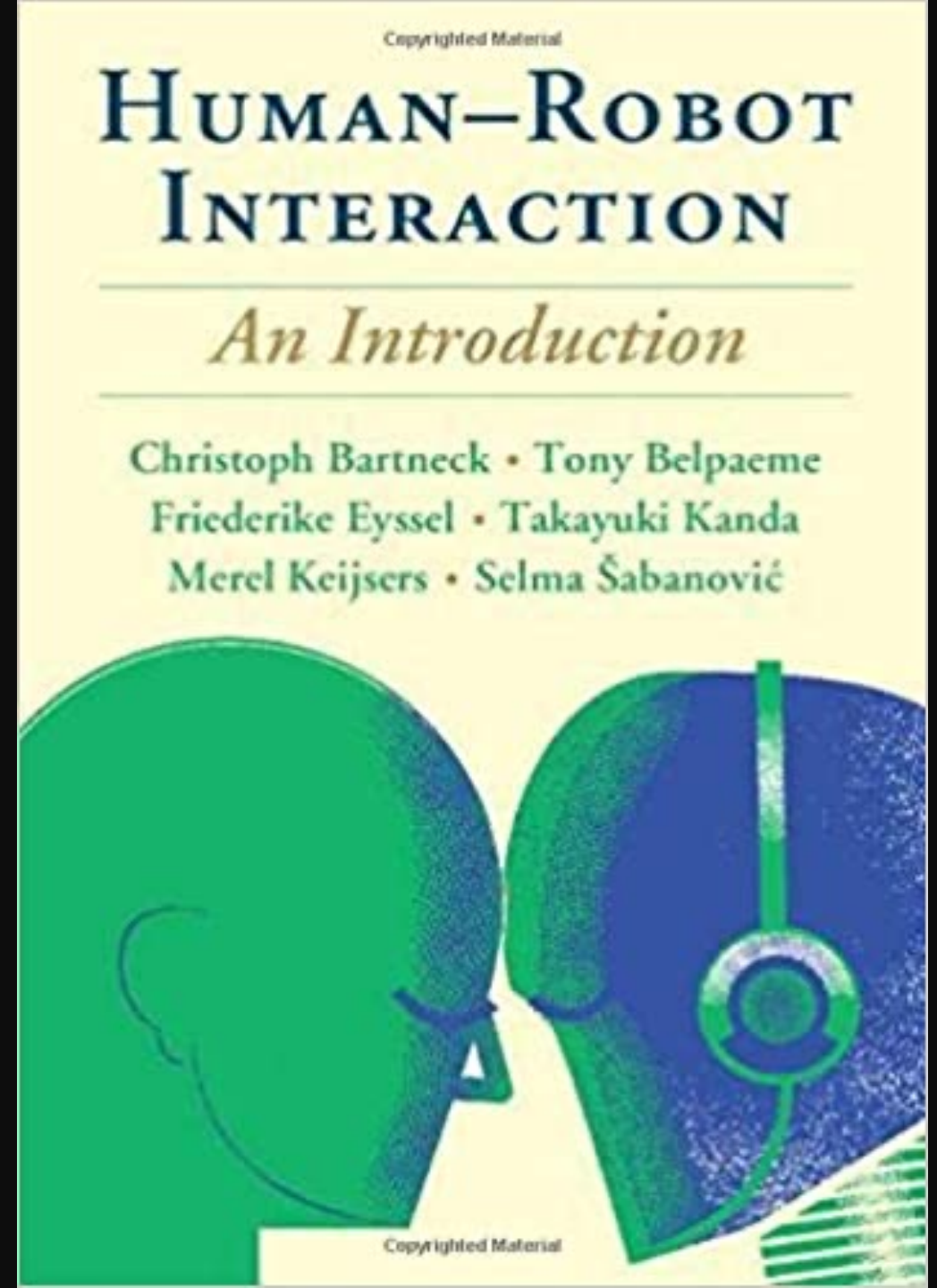
- Defining Research Question and Approach, Different Methods of Research, Choosing a Robot and Setting up the Mode of Interaction, Selecting Appropriate HRI Measures, Research Standards

Society and Ethics in HRI

- Robot in Society, Robots in Popular Media, Ethical Concerns in HRI, Future of HRI.

Text Books

- Bartneck, C., Belpaeme, T., Eyssel, F., Kanda, T., Keijsers, M. and Šabanović, S., 2020. **Human-robot interaction: An introduction**. Cambridge University Press.
 - Introduction to Robotics- John J. Craig, Third edition, Prentice Hill.
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Marks distribution

Segments	Marks (100)
Attendance	= 5
In-course examination	= 25
Final examination	= 70
Total Marks	= 100

Attendance (%)	Marks (05)
90 and above	05
85-89	04
80-84	03
75-79	02
60-74	01
Less than 60	00

What is HRI?

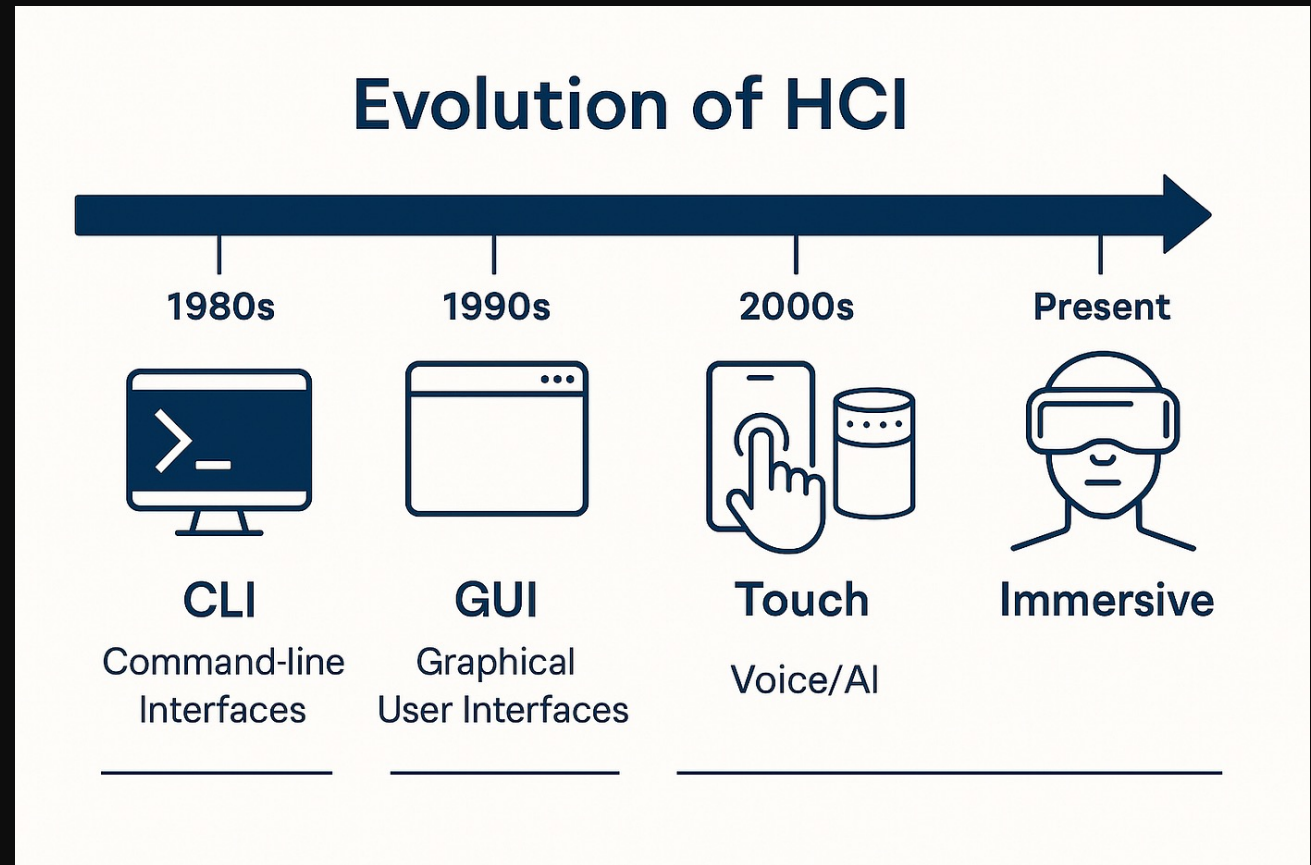
- Human–robot interaction (HRI) is the study of **interactions** between humans and robots.
 - It is a **multidisciplinary field** with contributions from human–computer interaction (HCI), robotics, artificial intelligence, natural-language understanding, design engineering, and psychology.
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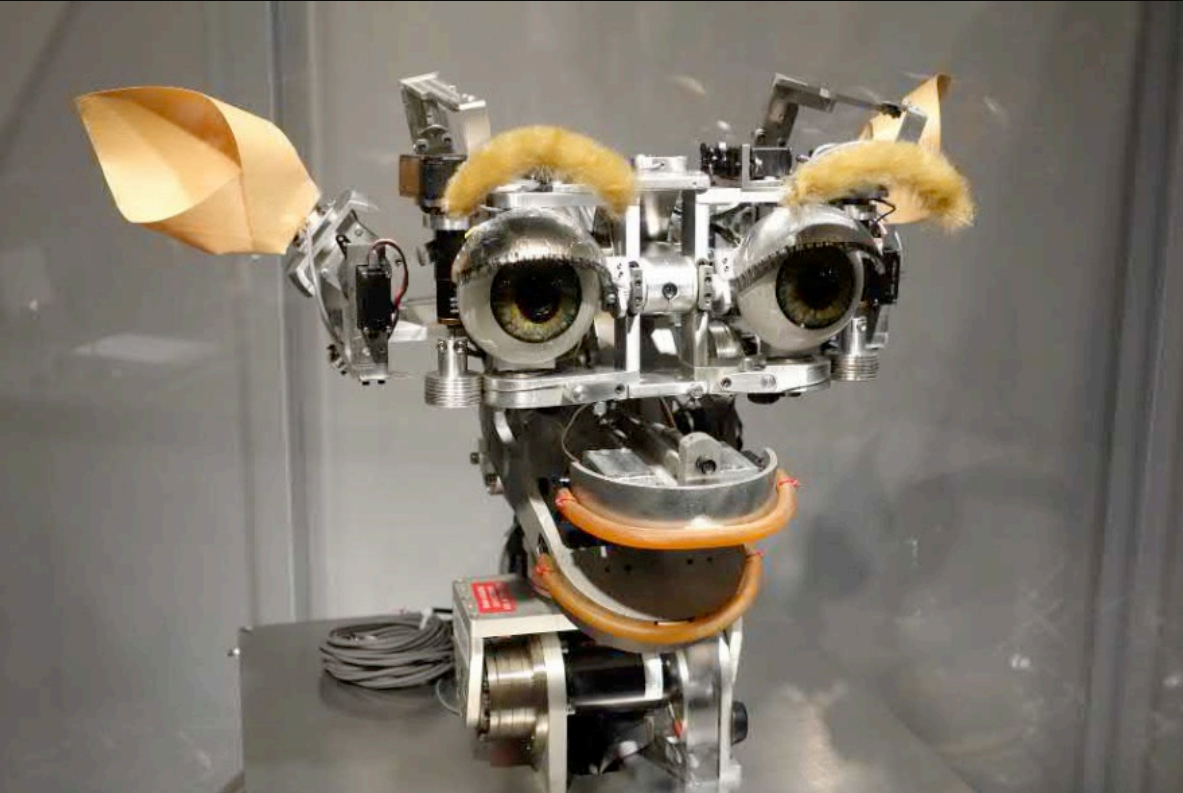
HRI vs. Robotics



HRI vs. HCI



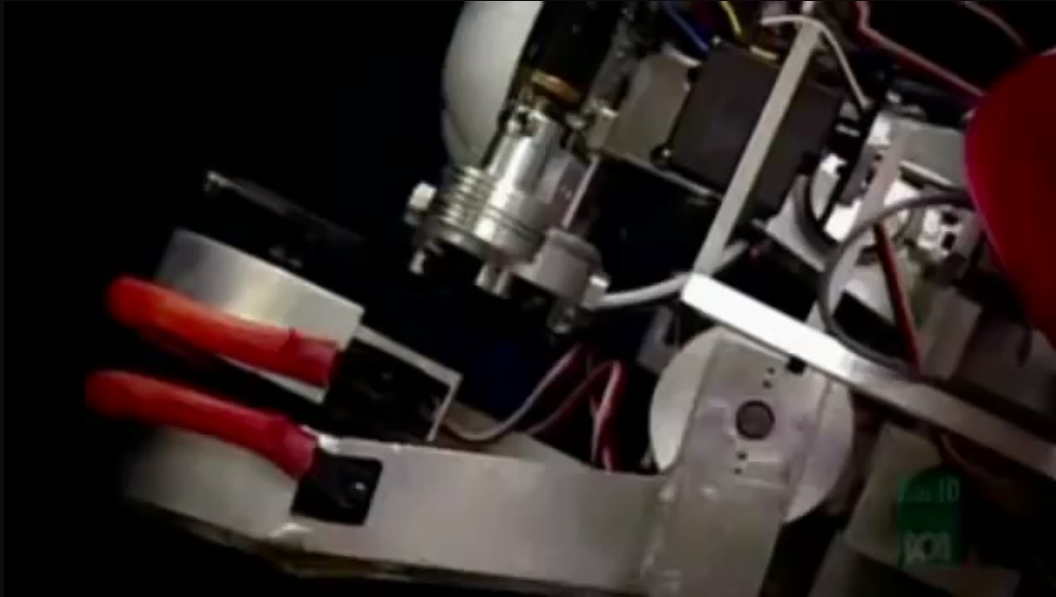
The HRI field draws insights from many other fields, but what other fields could benefit from research in HRI?



Kismet

Evolution of HRI

- The first mention of “**social robot**” in print was in 1935, when it was used as a derogatory term for a person having a cold and distant personality.
 - In 1978, the first mention of “social robot” was made in the context of robotics.
 - An early example of a social robot is **Kismet**, which was developed at the Massachusetts Institute of Technology in **1997**.
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Kismet

Evolution of HRI

- Kismet could animate its eyes, eyebrows, lips, and neck.
 - Based on visual and auditory input, it reacted to objects and people appearing in its visual field.
 - It extracted information on visual motion, sound amplitude, and emotion from speech prosody.
 - It responded by animating its facial expressions, ears, and neck, and by babbling in a nonhuman language.
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Nao robot

Evolution of HRI

- The Nao robot, developed by Aldebaran Robotics, is perhaps the **most influential robot** in the study of social robotics
 - First sold in 2006, the small humanoid robot, due to its **affordability, robustness, and ease of programming**, became a widespread robot platform for studying HRI.
 - The robot, because of its size, is also **highly portable**, allowing for studies to be run outside the lab.
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Keepon Robot

Evolution of HRI

- The **Keepon robot**, developed by Hideki Kozima, is a **minimal robot** consisting of two soft yellow spheres to which a nose and two eyes are added.
 - The robot can swivel, bend, and bob, using motors worked into the base of the robot.
 - Keepon was later commercialized as **an affordable toy**, and through some moderate hacking, it can be used as a research tool for HRI.
 - Studies with the Keepon robot convincingly demonstrated that **a social robot does not need to appear humanlike**; the simple form of the robot is sufficient to achieve interaction outcomes.
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Evolution of HRI

- **The Paro companion and therapy robot**, shaped like a baby seal, has been particularly popular in the study of socially assistive robots in eldercare, as well as other scenarios.
 - Paro has been commercially available in Japan since 2006 and in the United States and Europe since 2009.
 - Paro has been used by various psychologists, anthropologists, and health researchers, both to study the **potential psychological and physiological effects** on people and to explore ways in which robots might be adopted in healthcare organizations.
 - At the same time, the fact that it is a **closed platform** that does not allow robot logs or sensor data to be extracted from the robot or allow the robot's behaviors to be changed poses some limitations for HRI research.
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The Paro

Evolution of HRI



Baxter robot

- The Baxter robot, sold by Rethink Robotics until 2018, is both an **industrial robot** and a platform for HRI.
 - In contrast to the stiff robot arms of typical industrial robots, Baxter's arms move in response to an externally applied force.
 - In combination with other safety features, the Baxter robot is **safe to work near**, which makes it suitable for collaborative tasks.
 - In addition, Baxter has a display screen mounted at head height on which the control software can **display facial animations**.
 - Baxter's face can be used to **communicate its internal state**, and its eye fixations **communicate a sense of attention** to the human co-worker.
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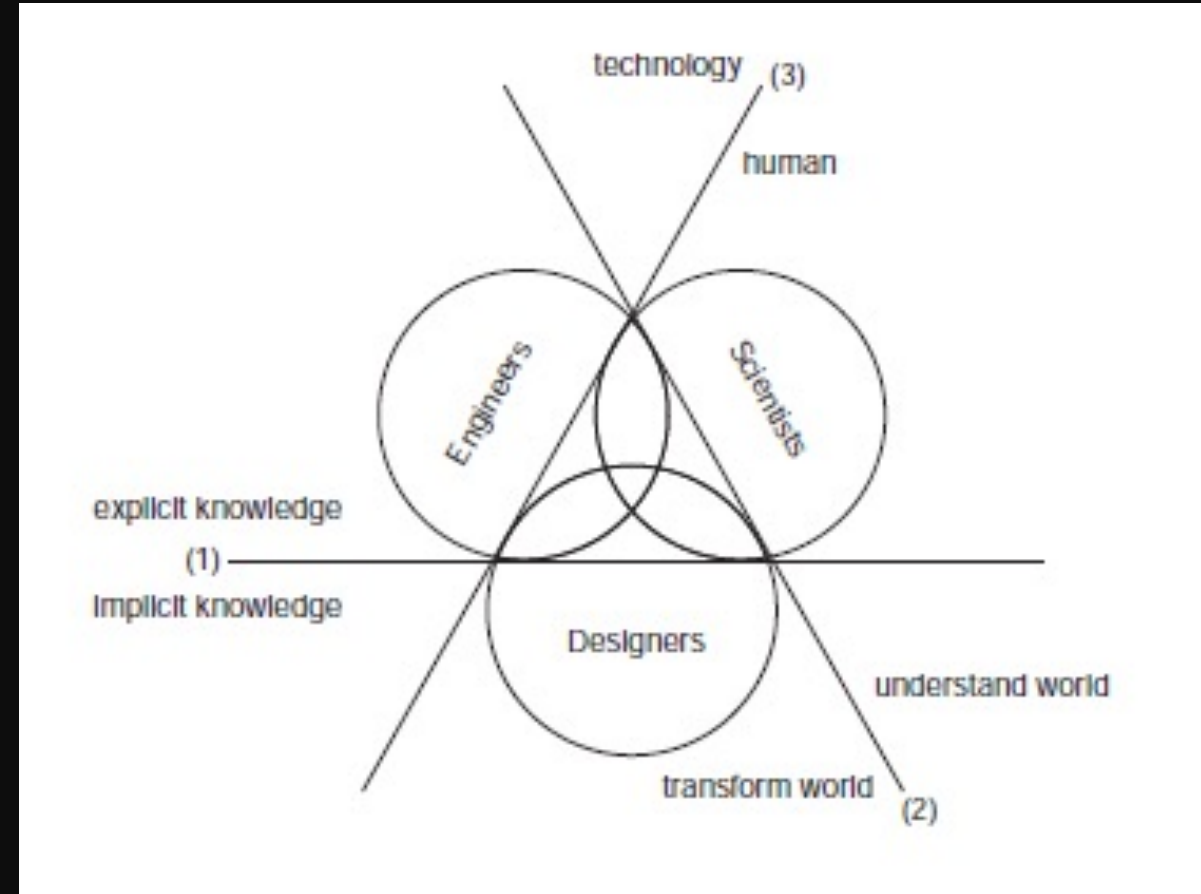
Challenges in HRI as an interdisciplinary endeavor

- Are you a designer, engineer, or social scientist? Try to imagine a situation in which you are collaborating with others to construct a robot (e.g., if you are an engineer, you are now working with a designer and a social scientist on this endeavor). How is your way of working different from the approaches that the other teammates might use?
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Challenges in HRI as an interdisciplinary endeavor

The difficulties of working together on a shared project find their base in three barriers between designers [D], engineers [E] and scientists (in particular social scientists) [S]:

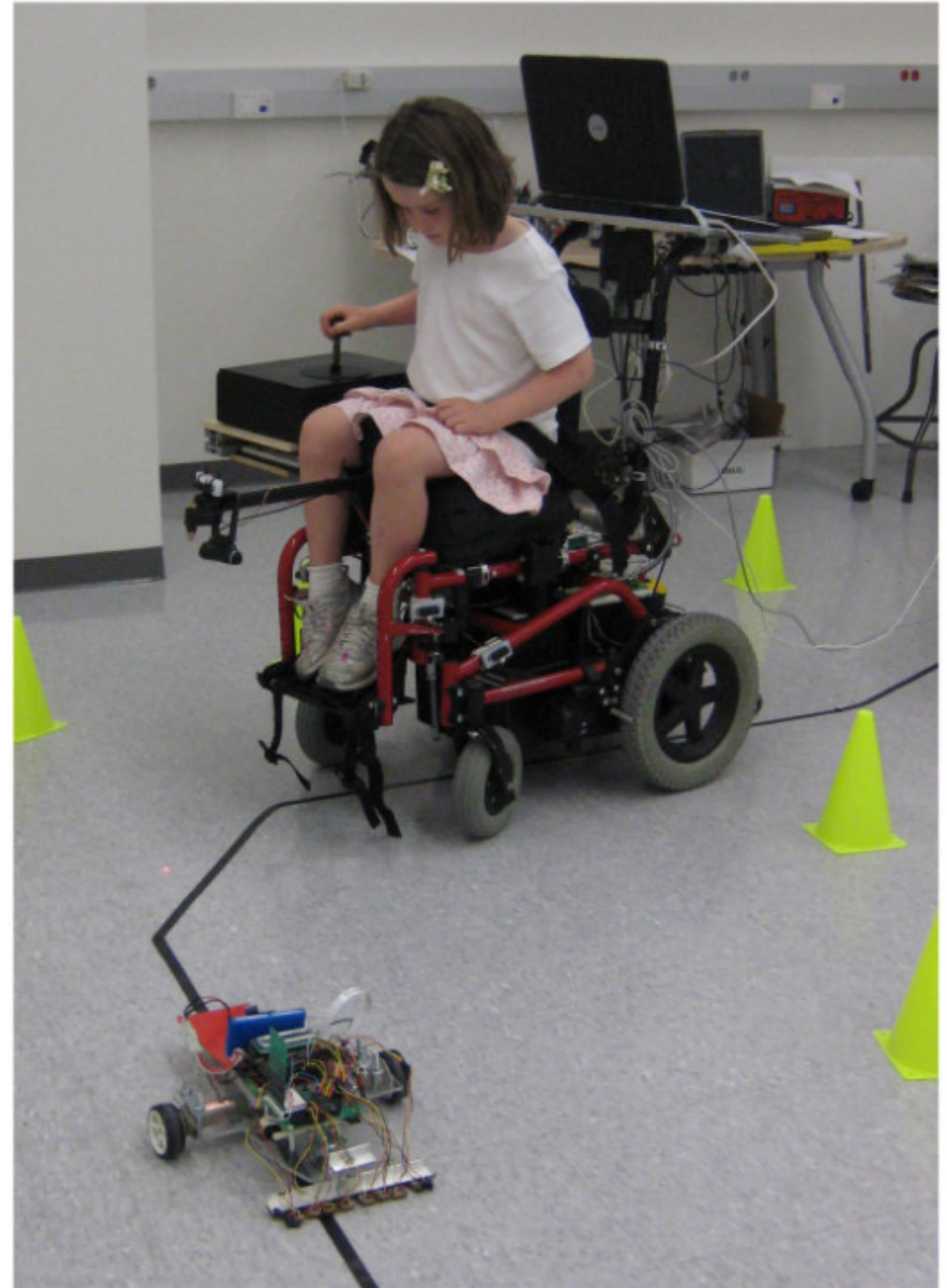
1. Knowledge representation (explicit [S, E] versus implicit [D]);
 2. View on reality (understanding [S] versus transforming reality [D, E]); and
 3. Main focus (technology [E] versus human [D, S]).
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Taxonomy for HRI

1. Autonomy level/ Amount of interaction

- In this category, we measure the autonomy level (**AUTONOMY**) and the amount of intervention required (**INTERVENTION**).
 - The **autonomy level** measures the percentage of time that the robot is carrying out its task on its own.
 - The amount of **intervention** required measures the percentage of time that a human operator must be controlling the robot.
 - These two measures **sum to 100%**.
 - Robots can have **full autonomy, full intervention, or shared control**.
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Taxonomy for HRI

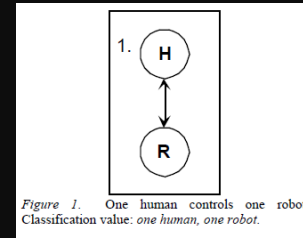
2. Ratio of people to robots

- The ratio of people to robots directly affects the human-robot interaction in a system.
 - The value of HUMAN-ROBOT-RATIO is denoted as a **non-reduced fraction**, with the number of humans over the number of robots.
 - If the number of operators or the number of robots is **variable** within the system, a range may be specified in either the numerator or the denominator of the ratio.
 - This taxonomy classification **does not measure the interaction** between the operators and the robots, simply the number of each.
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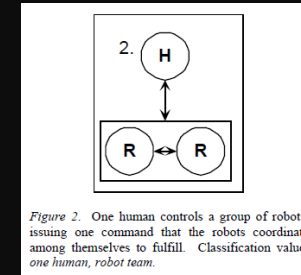
Taxonomy for HRI

3. Level of shared interaction among teams

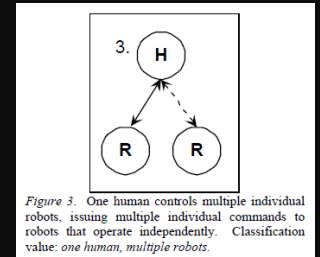
- If there are multiple human controllers, are these humans agreeing on commands before providing the robot(s) direction, or are they independently issuing commands that the robot(s) need to prioritize and/or deconflict?
- Also, if there are multiple robots, are they each receiving and acting on commands independently, or are all robots receiving all commands and coordinating among themselves to determine which robot(s) should respond to which commands?



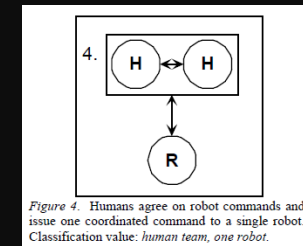
A person operating a wheelchair



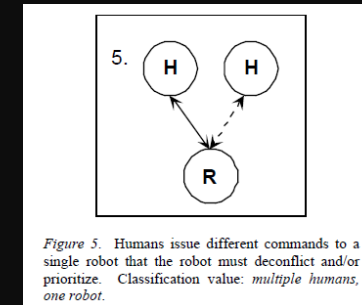
A person instructs a group of cleaning robots



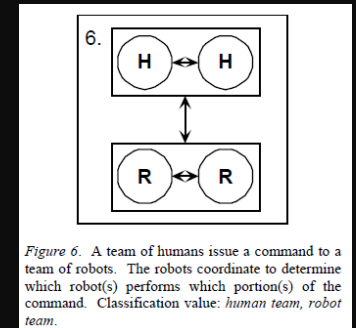
A person is engaged in multiple collaborations using an instant messenger



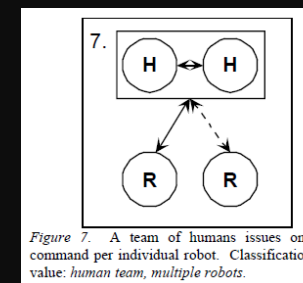
A firefighter and an emergency medical technician together agree on how a rescue robot should be directed



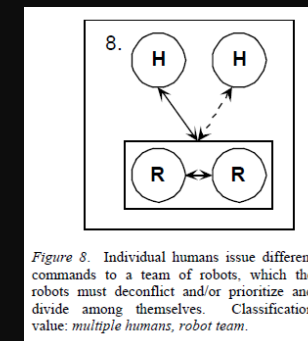
A delivery robot, who is asked to deliver a sandwich to one person and a steak to another



a search-and-rescue operation that requires multiple robots to carry out tasks



Multiple soldiers work together to direct individual robots to each destroy a different target



Individuals place orders to an online shop that handles orders using a coordinated robot team.

Taxonomy for HRI

4. Decision support for the operator:

- This taxonomy category has a few subcategories:
 - **Available sensor information:** The specification of available sensors is a list of sensing types available on the robot platform; the list may also contain the location of the sensors.
 - **Sensor information provided:** The sensor information provided, PROVIDEDSENSORS, is also a list of sensing types, which is a subset of AVAILABLE-SENSORS. All of the available sensor data may not be required for decision support.
 - **Type of sensor fusion:** Specified as a list of functions. For example, if sonar and lidar values were used to build a map that was displayed, the sensor fusion list would contain {{sonar, lidar}→map}.
 - **Pre-processing:** Finally, the amount of pre-processing of sensors for decision support is denoted in the PRE-PROCESSING list. If a video stream is processed prior to display to highlight regions of a particular color, say red, the list would include {video → highlight red regions}.
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Taxonomy for HRI

5. Criticality

- Criticality measures the importance of getting the task done correctly in terms of its negative effects should problems occur.
- Due to its subjective nature, CRITICALITY is broken into three categories: **high**, **medium**, and **low**.
- Urban search and rescue has **CRITICALITY=high**; it is dangerous for its user to be near the disaster situation, and it is important to find survivors quickly without damaging the building or hurting a trapped person.
- Robot soccer has **CRITICALITY=low**; if a robot team fails while playing another robot team, it will simply lose the game.
- The hospital delivery robot is an example of **CRITICALITY=medium**.



Taxonomy for HRI

6. Time/space taxonomy:

- The time-space taxonomy divides HRI into four categories based on whether the **humans and robots are using computing systems** at the same time (synchronous) or different times (asynchronous), and while in the same place (collocated) or in different places (non-collocated).

		Time	
		Same	Different
Space	Same	Robot Wheelchair	Manufacturing robots
	Different	Urban Search and Rescue	Mars Rover

Table 1: Time-space taxonomy category, with examples.

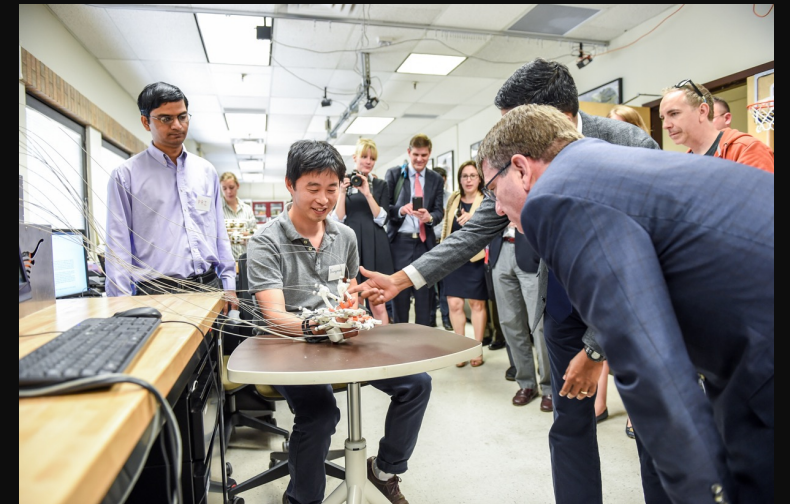
Taxonomy for HRI

7. Composition of Robot Teams

- A classification for denoting whether robot teams contain different types of robots or the same type of robot, if there is more than one robot in use.
 - This measure is called ROBOT-TEAM-COMPOSITION and can have the values **homogeneous or heterogeneous**.
 - Heterogeneous may be further specified with **a list containing the types of robots** in the team and the number of each type of robot used in the team.
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Application areas of HRI

1. **Industrial robots:** Industrial robots have been implemented to collaborate with humans to perform industrial manufacturing tasks.
2. **Medical robots:**
 - i. **Rehabilitation:** A rehabilitation robot is an example of a robot-aided system implemented in health care. This type of robot would aid stroke survivors or individuals with neurological impairment in recovering their hand and finger movements.
 - ii. **Elder care and companion robots:** Nursing robots are aimed to assist elderly people who may have faced a decline in physical and cognitive function, and, consequently, developed psychosocial issues.

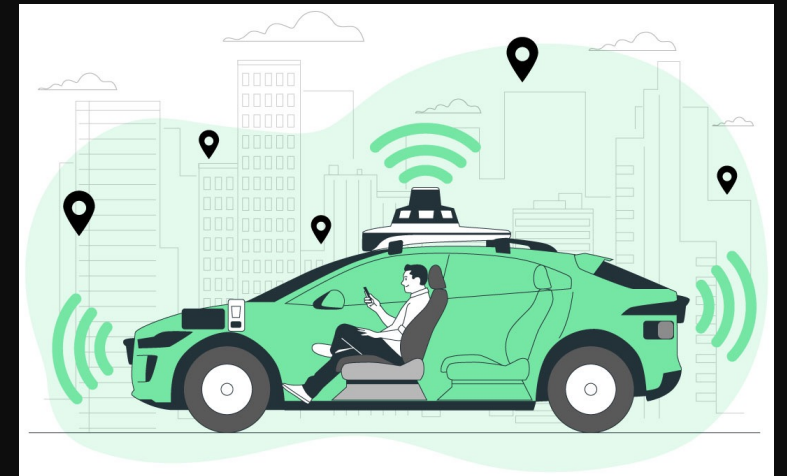


Application areas of HRI

3. Social robots:

- i. **Autism intervention:** Children with autism spectrum disorders (ASD) are more likely to connect with robots than humans, and using social robots is considered to be a beneficial approach to help these children.
- ii. **Education robot:** Robots can become tutors or peers in the classroom.

- 4. **Automatic driving:** A specific example of human–robot interaction is the human-vehicle interaction in automated driving. The goal of human-vehicle cooperation is to ensure safety, security, and comfort in automated driving systems.



Application areas of HRI

5. **Search and rescue:** Unmanned aerial vehicles (UAV) and unmanned underwater vehicles (UUV) have the potential to assist search and rescue work in wilderness areas, such as locating a missing person remotely from the evidence that they left in surrounding areas.
6. **Space exploration:** The collaboration between rovers, UAVs, and humans enables leveraging capabilities from all sides and optimizes task performance.
7. **Agricultural robots:** Human labor has been greatly used in agriculture but Agricultural robots like milking robots have been adopted in large-scale farming.





Major research areas in HRI

1. Methods for perceiving humans
 - Methods for perceiving humans in the environment are based on sensor information. This includes vision system, speech recognition system, natural language processing, etc.
2. Methods for motion planning
 - Motion planning in dynamic environments is a challenge that can at the moment only be achieved for robots with 3 to 10 degrees of freedom.





Major research areas in HRI

3. Cognitive models and the theory of mind
 - Humans exhibit negative social and emotional responses as well as decreased trust toward some robots that closely, but imperfectly, resemble humans; this phenomenon has been termed the "Uncanny Valley."
 - However, recent research in telepresence robots has established that mimicking human body postures and expressive gestures has made the robots likable and engaging in a remote setting.



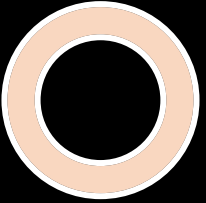


Major research areas in HRI

4. Methods for human–robot coordination

- A large body of work in the field of human–robot interaction has looked at how humans and robots may better collaborate.
- The primary social cue for humans while collaborating is the shared perception of an activity, to this end, researchers have investigated anticipatory robot control through various methods, including monitoring the behaviors of human partners using eye tracking, making inferences about human task intent, and proactive action on the part of the robot.





Q & A

